

City Simulation Project Plan

1. Project Overview

- **Goal:** Create a simplified city simulation starting with 10,000 people.
- **Representation:** Color-coded 12×12 grid.
- **Focus:** Population growth, aging, housing, employment, schools, and water.
- **Scope:** Simple rules, readable code, completion within a few days.

2. City Grid

- **Grid:** 12×12 (144 total blocks).
- **Starting Distribution:** 25 Residential, 20 Workplace, 25 Water, 5 Schools, 69 Empty.
- **Roads:** Assumed to exist; no separate blocks required.

3. Block Types

- **Enum:**

CSharp

```
enum TileType { Residential, Workplace, School, Water, Empty }
```

- **Class Tile:**

CSharp

```
class Tile {  
    public int X, Y;  
    public TileType Type;  
    public int Capacity;  
    public bool HasWater;  
    public List<Person> Residents, Workers, Students;
```

```
}
```

4. Starting Population

- **Total:** 10,000 people.
- **Residential:** 25 blocks, each starting with 400 residents (Capacity: 425).

5. Person Model

- **Class Person:**

CSharp

```
class Person {  
    public int Age;  
    public Tile HomeBlock, WorkBlock, SchoolBlock;  
    public bool IsEmployed;  
}
```

6. Initial Age Distribution

- 0–17 (20%): 2,000
- 18–40 (35%): 3,500
- 41–65 (30%): 3,000
- 66+ (15%): 1,500

7. Age Categories

- **Children:** 0–4.
- **Students:** 5–17.
- **Working-Age:** 18–65.
- **Retired:** 66+.

8. Aging

- `Person.Age += 1` every simulation year.

9. Birth System

- **Formula:** `Births = (People ages 22-40) * 0.02`.
- **New citizens:** Age 0.

10. Death System

- **Age-based probability check per year:**
 - 0-49: 0.1% | 50-65: 0.5% | 66-79: 2% | 80-89: 8% | 90+: 20%.

11. Residential Housing

- **Capacity:** 425 per block.
- **Shortage:** `Population > Capacity` → Convert Empty to Residential.

12. Workplace System

- **Capacity:** 300 jobs per block.
- **Assignment:** Eligible adults (18-65). Unemployment is allowed.

13. School System

- **Capacity:** 500 students per block.
- **Assignment:** Mandatory for 5-17.
- **Expansion:** `Students > Capacity` → Convert Empty to School.

14. Water System

- All developed blocks (Residential, Workplace, School) automatically have water.

15. Mayor Class

- **Responsibilities:** Assign jobs/schools, monitor capacity, convert empty blocks.

16. City Class

- Contains grid, population list, current year, and Mayor instance.

17. Simulation Class

- Logic to advance time.

18. Simulation Flow (Simulate Year)

- 1. Increase Year.
- 2. Age Population.
- 3. Handle Deaths.
- 4. Handle Births.
- 5. Check Housing/Schools (Build if needed).
- 6. Assign Students.
- 7. Assign Jobs.
- 8. Update UI.

19. User Interface

- **Statistics:** Year, Population, Employed, Block counts.
- **Grid:** 12×12 color-coded visualization.
- **Details Panel:** Click to view specific block info.

20. Development Stages

- 1. Data Model.
- 2. Population Logic (Aging/Births/Deaths).
- 3. Mayor logic.
- 4. Visualization.

- 5. Simulation integration.

6. Testing.